



Multi-Agent Debate with Reinforcement Learning for Faithful Hallucination Detection in Clinical Question Answering over Electronic Health Records

Multi-agent debate + MARL-trained verifier/challenger agents

Clinicians need fast, trustworthy answers to patient-specific questions over Electronic Health Records — but frontier LLMs hallucinate on real charts, and patient data cannot leave the hospital to use them.

Why this is hard

Hallucination is unsafe.

One-shot LLMs invent drugs, quote wrong lab values, and treat discontinued medications as current. In clinical use, these are patient-safety incidents — not minor errors.

Frontier LLMs are not deployable on PHI.

Sending a FHIR chart to GPT-5, Claude, or Gemini violates HIPAA data-residency requirements. The system must run on infrastructure the hospital controls.

One patient generates a huge amount of data — and it must never mix with another patient's.

A single chart holds thousands of records across notes, labs, meds, and encounters. The system has to handle that scale per patient while guaranteeing that one patient's data can never leak into another's answer.

What the system must do

1 Answer patient-specific clinical questions grounded in the patient's own chart

3 Refuse — explicitly — when evidence is insufficient

2 Provide a citation for every claim

4 Run locally, with no patient data sent to external APIs

Develop a cooperative Multi-Agent Reinforcement Learning (MARL) framework over a specialized multi-agent clinical reasoning system — producing faithful, evidence-grounded responses from a patient's chart, with explicit abstention when evidence is insufficient, running entirely on hospital-controlled infrastructure.

Train verifier/challenger policies with GRPO and MARL to improve hallucination detection under aligned prompts.

1 Train cooperating agents with MARL.

2 Ground every claim in the patient's own chart.

3 Provide per-claim traceability.

4 Refuse when evidence is insufficient.

5 Guarantee per-patient isolation.

6 Hospital-controlled inference.

7 Outperform frontier LLMs on real charts.

8 Train verifier/challenger policies with GRPO and MARL to improve hallucination detection under aligned prompts.

Novel Contributions

1

Multi-Agent Debate (MAD) — a bias-controlled verification architecture for clinical hallucination detection.

We replace single-critic verification with a three-agent debate — Verifier, Challenger, Blind Judge — operating at the claim level.

2

Cooperative MARL with counterfactual credit.

We adapt counterfactual credit assignment to cooperating LLM agents: each agent is rewarded for how much its presence changes the final verdict, not for its local output.

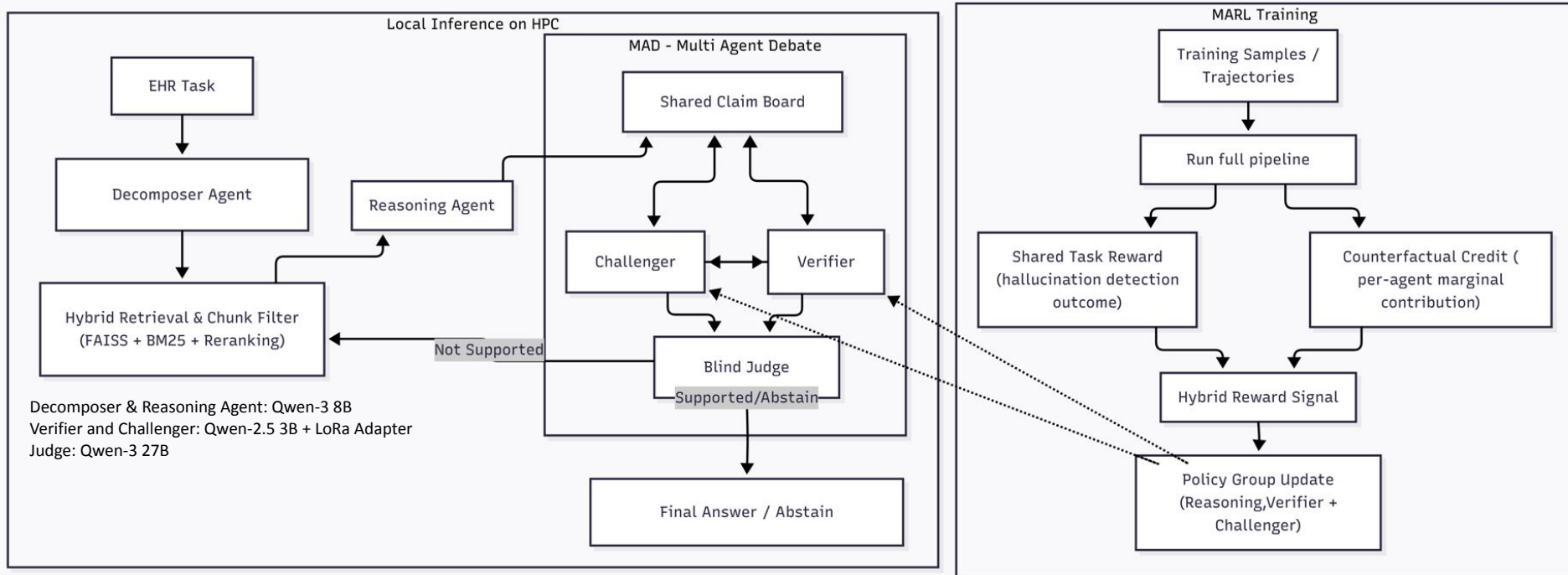
3

Fully on-prem, HIPAA-friendly deployment — runs on a single 80 GB GPU using open-source Qwen 3.

The entire system — retrieval, reasoning, debate, and MARL-trained adapters — runs on open-source Qwen models and fits on a single 80 GB H100. No external API calls, no PHI leaving the local environment, no third-party LLM dependency. Per-patient ephemeral indexes make cross-patient leakage impossible by construction. Privacy is enforced architecturally, not by policy.

Architecture Diagram

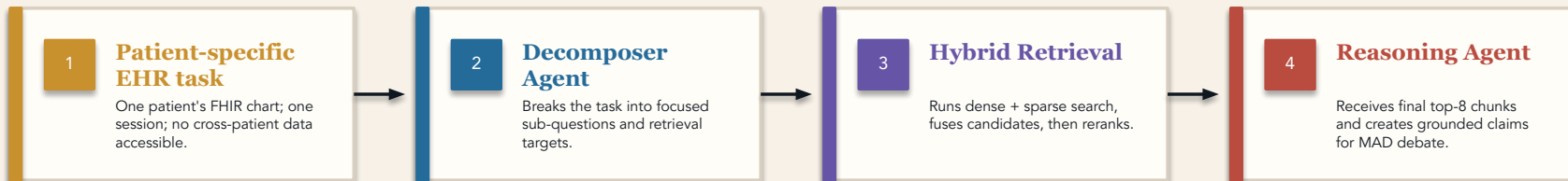
EHR System



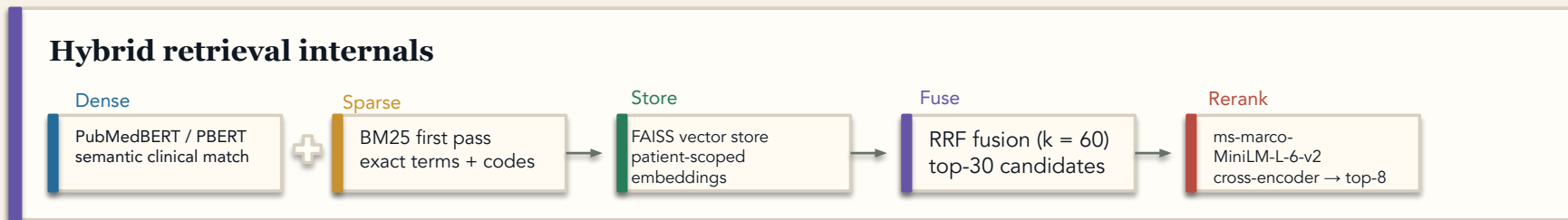
Build a patient-grounded evidence package locally, before reasoning.

The intake path is now four steps: EHR task, decomposition, hybrid retrieval, then reasoning over ranked evidence.

Four-stage intake path



Hybrid retrieval internals



Locality guarantee

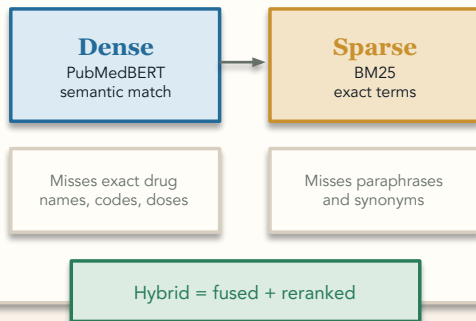
The model never answers from raw EHR text; it reasons only over a ranked, patient-grounded evidence package.

Why hybrid — not dense or sparse alone.

Dense retrieval finds semantic equivalents; sparse retrieval protects exact clinical terms. The system ships both, fused.

Two retrievers → one package

Dense and sparse fail differently, so the pipeline keeps both signals before reranking.

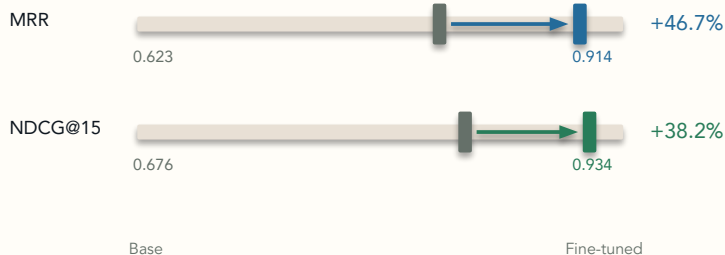


Runs locally

The final top 8 chunks are delivered to Reasoning agent Qwen 3 8B locally; no external API is needed.

Rank quality improves

Fine-tuning and reranking move the right chunks higher before generation.



1.000

Recall@15 saturated

Interpretation

The gain is ranking quality: better evidence reaches the reasoner earlier.

The only stage that writes: structured, cited drafts grounded in retrieved evidence.

Single LLM call · structured XML output · fault-tolerant parsing · runs locally on Qwen 3 8B.

Input it receives

- Patient-specific EHR task
- Top-8 chunks from hybrid retrieval
- Query type + branch flags from Decomposer
- Reasoning instructions

Output it produces

- `<reasoning>` step-by-step trace
- `<answer>` clinician-facing text with [1][2] citations
- `<source_chunks>` list of chunk IDs used
- DraftAnswer object passed downstream

How it works

Reads only top-8 chunks

The model has no access to the rest of the chart, preventing parametric knowledge bleed.

Reasons before answering

A structured reasoning trace is captured before the clinician-facing answer is emitted.

Outputs traceable claims

Every fact is anchored to one or more chunk IDs with inline [n] citations.

Model

Qwen 3 8B local

Temperature

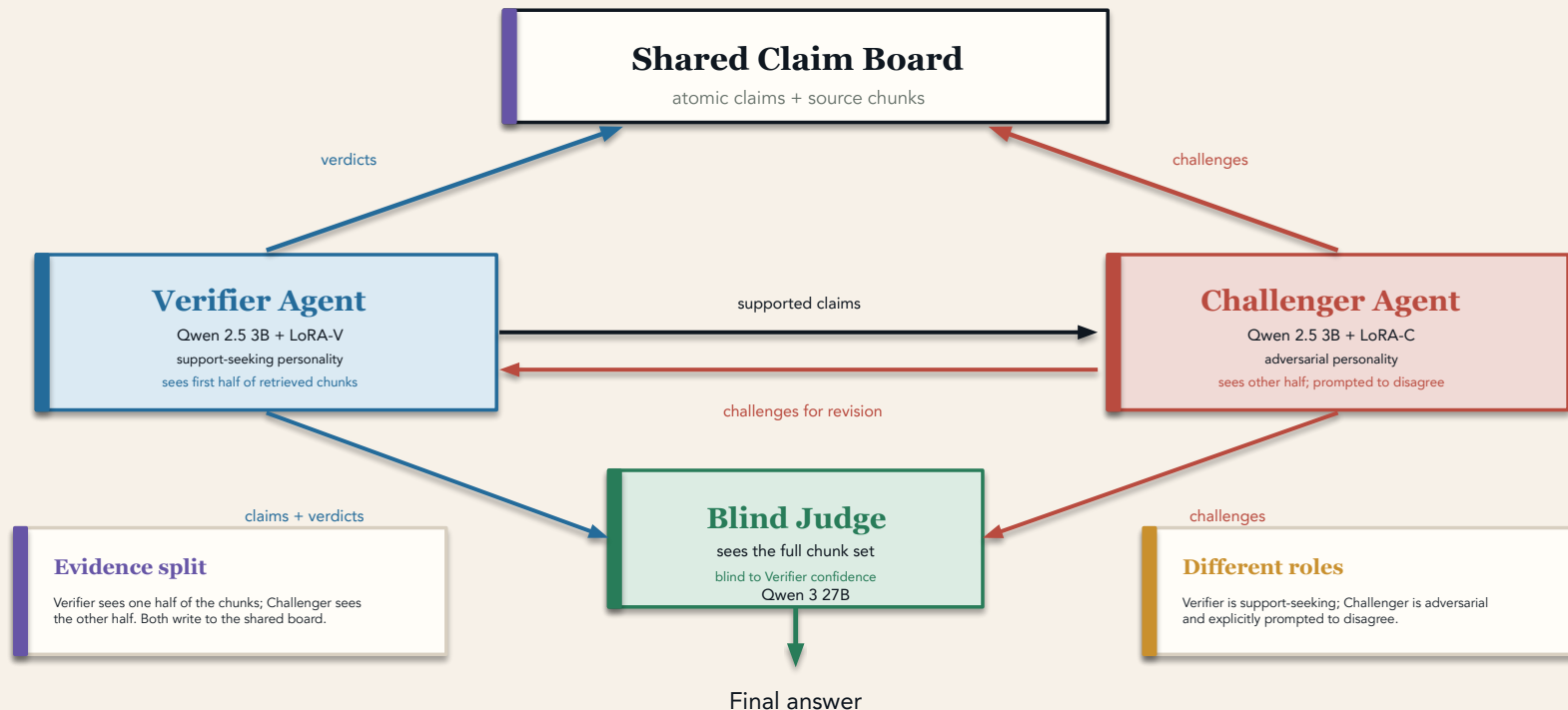
0.1 near-greedy

Latency

~1.5 s

A three-agent adversarial debate over a shared claim board — bias-controlled by construction.

Verifier supports · Challenger attacks · Blind Judge scores — no single agent has the final say.



A single critic checks; a debate creates pressure to find patient-specific failure modes.

The Challenger is not another reviewer. It is an adversary trained to expose unsupported claims.

Baseline

Single critic

One review pass; no forced counterexample search.

MAD debate

Verifier supports, Challenger attacks, Blind Judge scores.

+70% 0.463 → 0.789
recall lift

Benchmark movement

Single critic vs MAD debate on the same full-pipeline benchmark.

F1 overall detection improves



+17%

Precision precision tradeoff



+20%

Recall more hallucinations caught



+70%

Useful tradeoff

MAD catches far more misses, while precision falls modestly.

Four pieces make MARL work. Take one away, it collapses.

Independent training makes agents individually strong, but they do not cooperate. C3 v2 fixes that.

1

Counterfactual Credit

Reward each agent by how much its presence changes the team's result — not its own output.

Why: otherwise agents get better alone but do not cooperate.

2

Hybrid Reward

Mix three signals so no one dominates: individual correctness + counterfactual + team outcome.

Why: one signal alone is unstable or selfish.

3

Separate LoRA per Agent

Verifier and Challenger have their own weights. No sharing.

Why: shared LoRA failed 4 times — one agent erased the other's learning.

4

IBR + Warm-Start

Freeze one, train the other, swap, repeat — start from GRPO checkpoints.

Why: otherwise both agents move at once and never converge.

Reward formula

$$r = 0.5 r_{\text{individual}} + 0.3 r_{\text{counterfactual}} + 0.2 r_{\text{shared}}$$

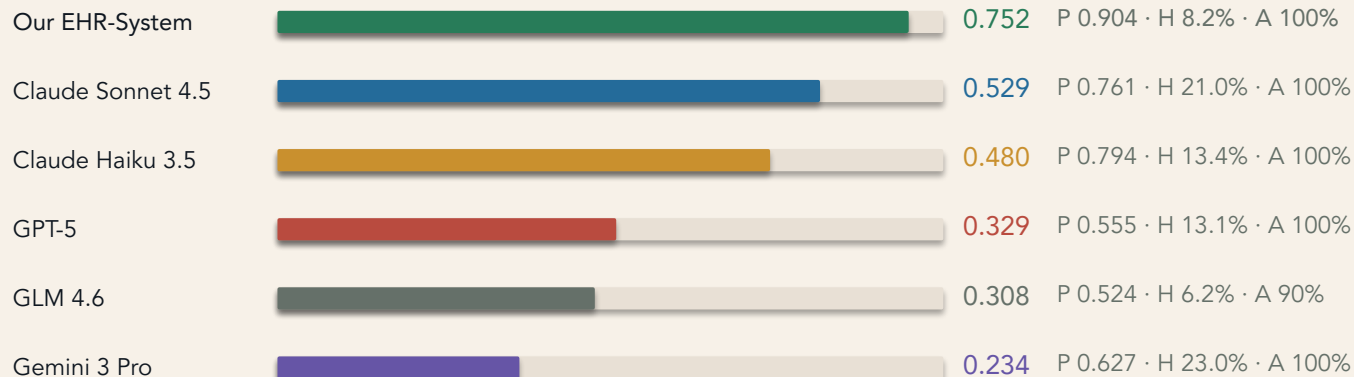
0.5 did you do your own job? · 0.3 did your presence matter? · 0.2 did the team get it right?

MARL C3 v2 Ablation: each improvement, what it fixes, and what breaks without it

Piece	What it is for	What it fixes	Without it	What improved	Final evidence
Independent GRPO v3	Warm-starts agents with hallucination-detection skill	Solves cold-start policy training; no critic network needed	Agents get better in isolation but do not coordinate	Verifier reward 0.221 → 0.705; Challenger 0.132 → 0.218; pipeline F1 only 0.642 → 0.657	Useful warm-start, not enough alone
Counterfactual credit	Rewards each agent's marginal contribution to the team	Individual rewards do not reward cooperation	A strong isolated agent may add no team value	Measures pipeline outcome with agent replaced by neutral default	Forces team usefulness
Hybrid reward $\alpha=.5$ $\beta=.3$ $\gamma=.2$	Blends individual skill, counterfactual credit, and shared task reward	Pure individual ignores team; pure counterfactual is noisy; pure shared free-rides	Training becomes unstable or does not coordinate	Stable reward: role skill + marginal lift + team verdict	Coordination signal
Separate LoRA per agent	Keeps Verifier and Challenger in separate parameter spaces	Shared LoRA gradients overwrite each other	Collapse / oscillation: shared-LoRA attempts fail around 43–49%	Four shared-LoRA attempts failed; separate adapters reached 82% accuracy	Make-or-break fix
IBR + warm-start	Freezes one partner while training the other from GRPO v3	Simultaneous MARL is non-stationary; scratch starts too slow	Target keeps moving; training stalls or misses compute budget	47% → 60.5% → 72% → 82%; matched F1 0.384 → 0.694	Monotonic MARL improvement
C3 v2 combined	All four fixes together	Credit assignment + reward variance + adapter interference + non-stationarity	Any missing piece breaks convergence or limits to no pipeline gain	Matched-prompt F1 0.752	Final recipe that works

Beats every frontier baseline on real charts.

Entity F1 ranking



+17%
over strongest
frontier baseline

0.904
precision

100%
abstention
accuracy

Bottom strip

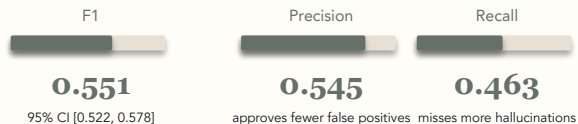
+17% over the strongest frontier baseline · only system with 100% abstention accuracy at top precision

MAD debate changes the error profile before any training.

Architecture-only ablation on 1,000 MedHallu pairs with 10,000-sample bootstrap CIs.

Single Critic

One reviewer checks the answer once. No forced counterexample search.

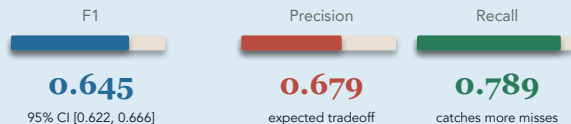


Failure mode

subtle supported-looking false claims pass through

MAD Debate

Verifier supports, Challenger attacks, Blind Judge scores without confidence anchoring.



Architecture effect

+0.094 F1 and +70% recall lift with no training

Bootstrap confidence

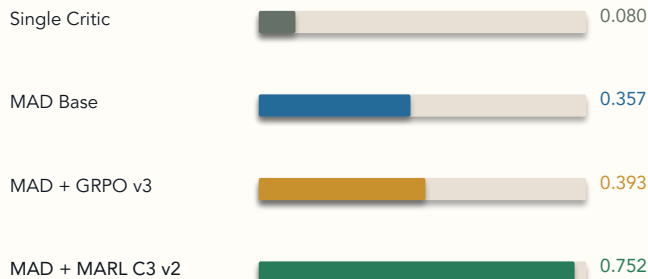
Single Critic CI [0.522, 0.578] and MAD Debate CI [0.622, 0.666] do not overlap.

Interpretation: the debate structure itself creates the gain.

MARL pushes detection F1 to 0.752 — and surfaces a new research problem.

Two columns side by side.

Matched-prompt evaluation



Research problem surfaced

MARL can train better debate agents under matched evaluation.

The remaining problem is prompt-distribution alignment: training prompts and deployment prompts must match closely enough for the learned policy to transfer.

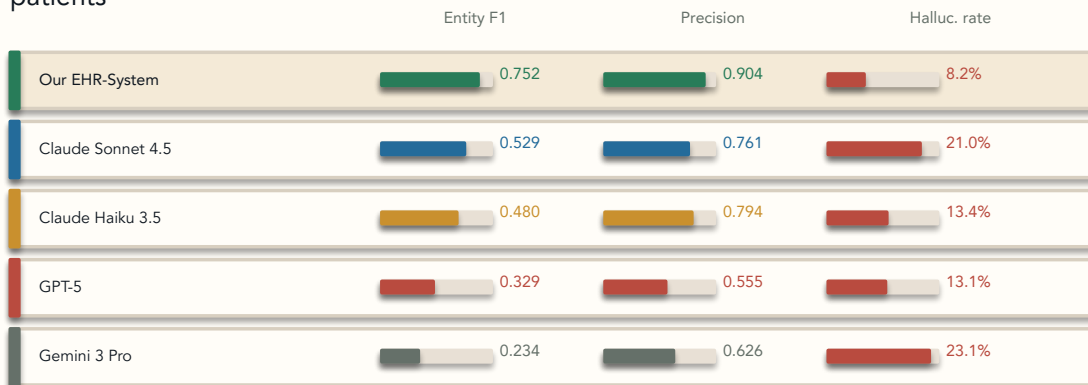
0.752
F1
matched

C3 v2
best recipe
separate LoRA per agent

Our EHR-System improves patient-specific entity accuracy.

It beats frontier LLM baselines on 78 clinical queries across 10 MIMIC-IV FHIR patients.

Pipeline benchmark: 78 queries / 10 MIMIC-IV FHIR patients



0.904

Precision

fewer unsupported entities

8.2%

Hallucination rate

lower is better

0.752

Medication F1

vs Sonnet 0.383

What it signifies: architecture-level controls improve factual grounding beyond general LLM ability.

Future Work

Self-improving debate with human-in-the-loop feedback: every clinician interaction becomes a training signal. Approvals reinforce the current debate behavior; corrections, edits, and overrides become negative examples that fine-tune the Verifier and Challenger continuously. The system gets better the more clinicians use it and learns the specific failure modes of this hospital's chart style, not a generic benchmark.

Partner with a hospital for prospective clinical deployment — move from MIMIC-IV data to a real chart-review workflow with practicing clinicians. Measure clinical impact directly:

Federated MARL across multiple hospitals — train debate agents collaboratively across institutions without ever moving PHI. Each hospital trains locally on its own charts; only the LoRA gradients are shared and aggregated.

References

- [1] DeepSeek-AI, "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning," Nature, vol. 645, pp. 633–638, Sep. 2025.
- [2] S. Pandit, J. Xu, Y. Hu, et al., "MedHallu: A Comprehensive Benchmark for Detecting Medical Hallucinations in Large Language Models," in Findings of the Association for Computational Linguistics: ACL 2025, Vienna, Austria, 2025.
- [3] J. Chen, Z. Cai, K. Ji, et al., "HuatuoGPT-o1: Towards Medical Complex Reasoning with Large Language Models," in Proc. Annu. Meeting Assoc. Comput. Linguistics (ACL 2025), 2025.
- [4] S. Khan, et al., "Debating with More Persuasive LLMs Leads to More Truthful Answers," in Proc. 42nd Int. Conf. Mach. Learn. (ICML 2025), Vancouver, Canada, Jul. 2025.
- [5] Z. Wang, Y. Wang, Y. Zhang, et al., "MMOA-RAG: Multi-Module Joint Optimization Algorithm for Multi-Agent RAG," in Proc. AAAI Conf. Artif. Intell. (AAAI 2025), Philadelphia, PA, USA, Feb. 2025.

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Specific objectives

- | | | |
|---|---|---|
| 1 | Train cooperating agents with MARL. | A Verifier, a Challenger, and a Blind Judge are trained jointly with counterfactual credit assignment and iterated best response — so each agent learns to complement the others, not duplicate them. |
| 2 | Ground every claim in the patient's own chart. | All generated content is anchored to retrieved chart evidence — no parametric knowledge bleed. |
| 3 | Provide per-claim traceability. | Each statement maps to a specific chart entry, making clinician verification a single click. |
| 4 | Refuse when evidence is insufficient. | A principled abstention policy replaces fabrication. |
| 5 | Guarantee per-patient isolation. | Session-scoped, in-memory indexes destroyed on session end. Cross-patient leakage impossible by construction. |
| 6 | Hospital-controlled inference. | No PHI leaves the local environment. |
| 7 | Outperform frontier LLMs on real charts. | Evaluated against GPT-5, Claude Sonnet 4.5, Gemini 3 Pro, GLM 4.6, and Haiku 3.5 on MIMIC-IV FHIR data with entity-level ground truth — not LLM-as-judge. |
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Cooperative MARL with counterfactual credit — and the working recipe.

We adapt counterfactual credit assignment (COMA / Shapley) to cooperating LLM agents: each agent is rewarded for how much its presence changes the final verdict, not for its local output. After 4 failed attempts with shared adapters, we identify the configuration that actually works — separate LoRA per agent, warm-started from independent GRPO, trained under Iterated Best Response — reaching F1 = 0.752 under matched evaluation.

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Fully on-prem, HIPAA-friendly deployment — runs on a single 80 GB GPU using open-source Qwen 3.

The entire system — retrieval, reasoning, debate, and MARL-trained adapters — runs on open-source Qwen 3 and fits on a single 80 GB H100. No external API calls, no PHI leaving the local environment, no third-party LLM dependency. Per-patient ephemeral indexes make cross-patient leakage impossible by construction. Privacy is enforced architecturally, not by policy.

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# Current LLMs are not safe enough for patient-specific EHR QA.

The issue is not general medical knowledge; it is faithful use of one patient's record.

## Hallucinations are clinical risks

A plausible answer can fabricate a lab, dose, date, medication, or diagnosis.

## Patient data is not static context

EHR facts are longitudinal, sparse, numeric, and tied to a specific patient timeline.

## Generic LLM prompting is brittle

The same prompt cannot safely handle factual, temporal, numeric, medication, and comparison queries.

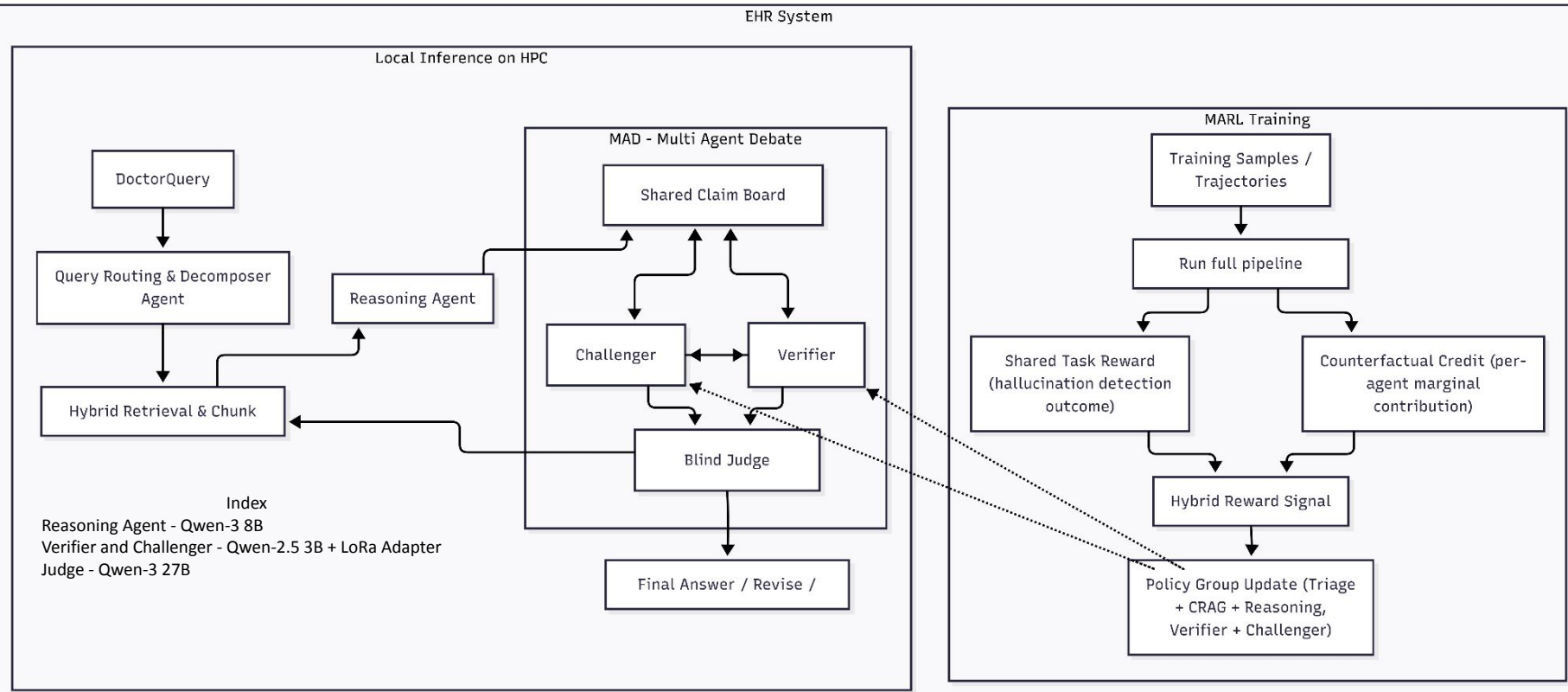
## Single critics miss subtle errors

One-pass verification lacks adversarial pressure and often misses supported-looking but false claims.

Core lesson

# Clinical QA needs an architecture that separates evidence retrieval, reasoning, adversarial verification, and final routing.

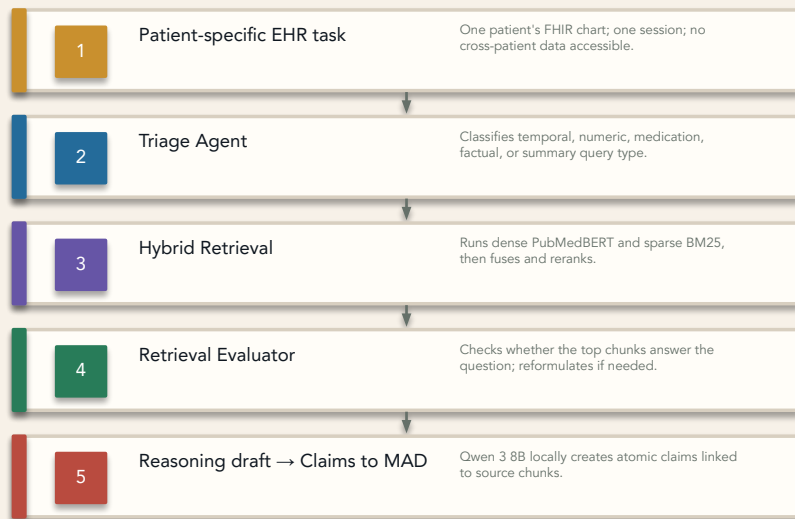
# Architecture Diagram



# Build a patient-grounded evidence package locally, before any reasoning happens.

Every query runs through hybrid retrieval. The model never answers from raw EHR text.

Five-stage intake path



## Hybrid Retrieval

always both, fused

Dense

Fine-tuned PubMedBERT finds semantic matches such as "hypertension" ≈ "high BP".

Sparse

BM25 preserves exact drug names, lab codes, ICD codes, and abbreviations.

Fuse

Reciprocal Rank Fusion merges both rankings ( $k = 60$ ).

Rerank

Cross-encoder rerank delivers the final top 8 chunks to the reasoner.

Locality guarantee

The model receives a ranked, patient-grounded evidence package before it is allowed to reason.

# Hybrid retrieval covers the failure modes of dense and sparse search.

Dense catches semantic equivalents; BM25 protects exact clinical terms; fused retrieval is what we ship.

Why hybrid — not dense or sparse alone

| Retriever            | Strength                     | Failure mode                          |
|----------------------|------------------------------|---------------------------------------|
| Dense (PubMedBERT)   | Catches semantic equivalents | Misses exact drug names, codes, doses |
| Sparse (BM25)        | Catches exact terms          | Misses paraphrases and synonyms       |
| Hybrid (both, fused) | Covers both                  | What we ship                          |

Key numbers

| Stage                                         | Result                                             |
|-----------------------------------------------|----------------------------------------------------|
| Fine-tuned PubMedBERT (760 clinical triplets) | MRR 0.623 → 0.914 (+46.7%)                         |
| Hybrid retrieval + rerank                     | Recall@15 = 1.000                                  |
| Reasoner                                      | Qwen 3 8B, runs on one 80 GB H100, no external API |

# Why hybrid — not dense or sparse alone.

Dense retrieval finds semantic equivalents; sparse retrieval protects exact clinical terms. The system ships both, fused.

Retriever tradeoff

## Dense PubMedBERT

Strength

Catches semantic  
equivalents

Failure mode

Misses exact drug  
names, codes, doses

## Sparse BM25

Strength

Catches exact terms

Failure mode

Misses paraphrases and  
synonyms

## Hybrid both, fused

Strength

Covers both

Failure mode

What we ship

Key numbers

**0.623 → 0.914**

MRR

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Hybrid retrieval + rerank

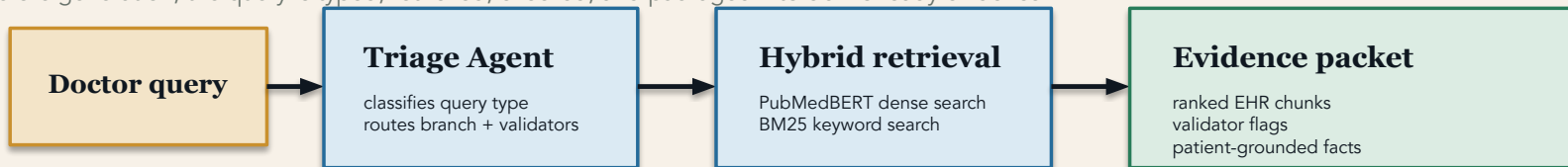
**Qwen 3 8B**

Reasoner

one 80 GB H100  
no external API

# The evidence package is the local, patient-specific context layer.

Before generation, the query is typed, retrieved, checked, and packaged into claims-ready evidence.



## PubMedBERT

Biomedical embedding model used to find semantically similar clinical evidence.

## BM25

Sparse keyword retriever that keeps exact terms, labs, drugs, and abbreviations visible.

## Chunks + flags

Chunks are retrieved EHR passages; flags mark numeric, temporal, and entity checks.

## Local LLM calls

Debate agents use Qwen 2.5 3B Instruct locally; reasoning uses MiniMax M2.5.

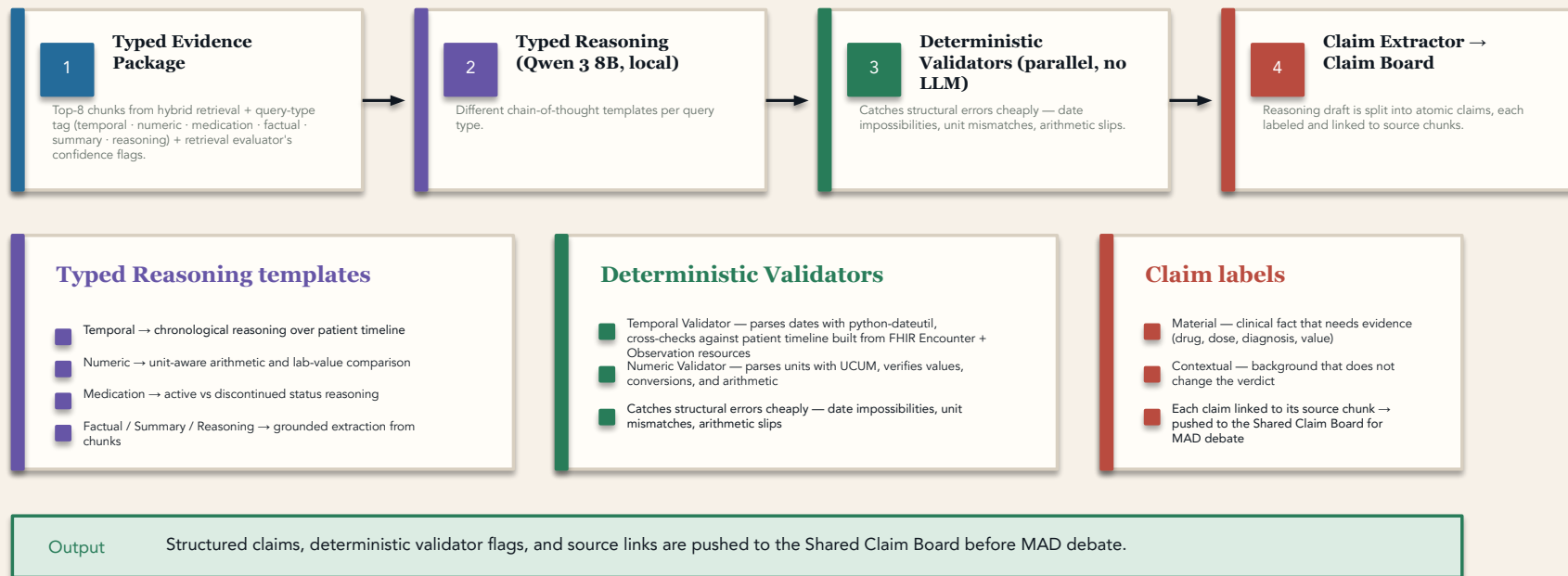
### Why this matters

The model does not answer from raw EHR text. It receives a typed, ranked, locally generated evidence packet.

# Reasoning runs after evidence is filtered, typed, and checked — and produces structured claims, not free prose.

Subtitle: Each query type activates a different reasoning template; deterministic validators catch structural errors before debate begins.

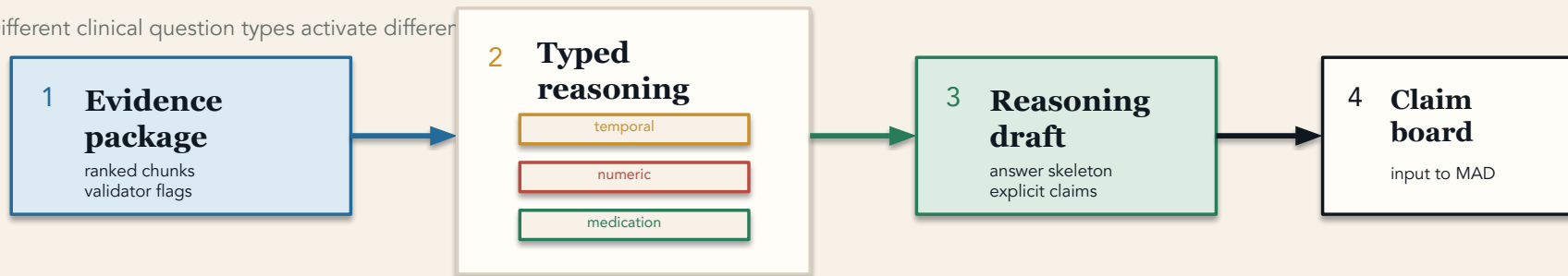
Flow (4 stages)





# Reasoning converts the evidence package into claim-producing drafts.

Different clinical question types activate different



## Design principle

The model reasons after evidence is filtered, typed, and checked.

## Safety principle

Numeric and temporal validators reduce errors before debate begins.

## Output principle

MAD receives atomic claims with supporting evidence, not a free-form blob.

# A three-agent adversarial debate over a shared claim board — bias-controlled by construction.

Subtitle: Verifier supports, Challenger attacks, Blind Judge routes. No single agent has the final say.

The three agents (label each box on the diagram with these)

1

## Verifier Agent (Qwen 2.5 3B + LoRA-V)

Scores each material claim under strict rules — supported only if the evidence explicitly states the same drug / dose / value. Marks ambiguous claims as low-confidence rather than approving them.

2

## Challenger Agent (Qwen 2.5 3B + LoRA-C)

Required to challenge every supported claim — agreement is structurally disallowed. Attacks using a fixed taxonomy:

CONTRAINDICATION · DOSAGE\_CHECK · INTERACTION · GUIDELINE\_CURRENCY · GAP\_FINDING

3

## Blind Judge Agent

Scores each claim independently, structurally blind to the Verifier's confidence — eliminates anchoring bias. Output is a ternary score per claim:

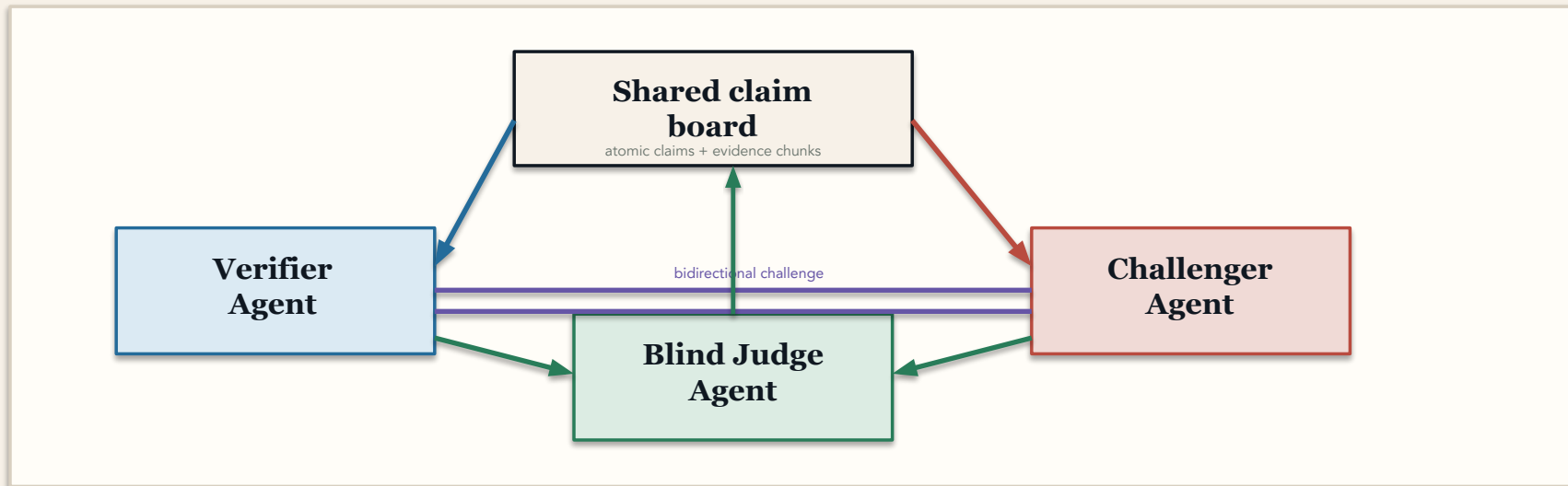
1.0 evidence explicitly supports · 0.5 topic match but specifics differ · 0.0 unsupported or fabricated

No single agent has the final say.

Verifier, Challenger, and Blind Judge each see a different role-specific view of the same claim board.

# MAD makes verification adversarial before an answer is released.

A shared claim board lets Verifier, Challenger, and Blind Judge work over the same patient-grounded state.



## Verifier Agent

Scores each material claim under strict evidence-matching rules.

## Blind Judge Agent

Audits SUPPORTED claims via a fixed taxonomy

## Challenger Agent

Scores each claim 1.0 / 0.5 / 0.0, blind to Verifier confidence

# Two structural anti-bias mechanisms — both essential.

Result — architecture alone, no training.

Mechanism

Why it matters

Challenger required to disagree

Prevents agreement collapse where both agents converge to "looks fine"

Judge blind to Verifier confidence

Eliminates anchoring bias documented in single-critic / CoT verification

Detection F1

**0.551** →  
on 1,000 MedHallu pairs  
**0.645**

**+17%**  
improvement  
non-overlapping  
95% bootstrap  
Cls.

**The biggest single gain in  
the system comes from this  
architectural change — not  
from scale, not from RL.**

Why this matters

A single critic checks. A debate creates pressure to find patient-specific failure modes. The Challenger is not another reviewer — it is an adversary trained to expose unsupported claims. The Judge is not just an aggregator — it is the only agent that sees both sides without confidence anchoring.

# Two structural anti-bias mechanisms — both essential.

Result — architecture alone, no training.

1

## Challenger must disagree

Prevents agreement collapse where both agents converge to "looks fine".

2

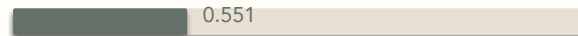
## Judge is blind to confidence

Eliminates anchoring bias from single-critic or chain-of-thought verification.

## Architecture ablation

Detection F1 on 1,000 MedHallu pairs · truncated axis 0.50–0.67

Single critic



MAD debate



0.50

0.67

+0.094

**+17%**  
architecture-only  
improvement

**+70%**  
reg lift

95% bootstrap  
CIs do not  
overlap.

### Design implication

The biggest single gain comes from the debate architecture itself: pressure to disagree plus a judge that is not anchored to verifier confidence.

# A single critic checks; a debate creates pressure to find patient-specific failure modes.

The Challenger is not another reviewer. It is an adversary trained to expose unsupported claims.

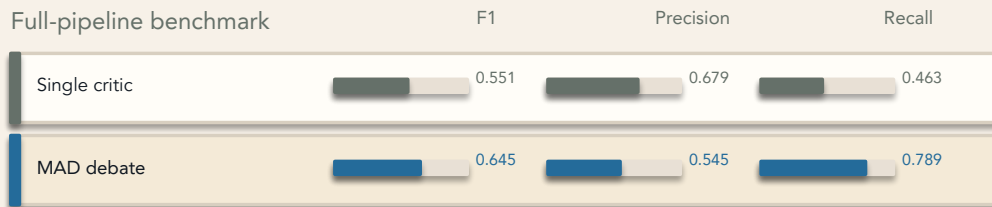
## Baseline: single critic

Answer is reviewed once, then accepted or rejected. There is no forced counterexample search.

## System: MAD debate

Verifier supports claims; Challenger attacks them; Blind Judge scores without confidence anchoring.

Full-pipeline benchmark



**+70%**

recall lift

0.463 -> 0.789

# Judge scores become reward signal; policy updates go back to Verifier and Challenger.

MARL is used to make agents cooperate, not to replace the evidence pipeline.

**0.752**

Matched-prompt F1  
MARL C3 v2

**0.643**

Full-pipeline F1  
transfer gap remains

**+111%**

Matched lift  
vs base MAD

## Lesson

Most useful when prompts match deployment.

# Each improvement maps to a different safety control.

The system is not one model trick; it is a layered architecture.

## Evidence package

MRR 0.623 -> 0.914  
Better rank quality before  
generation.

## MAD

F1 0.551 -> 0.645  
Recall +70% over single critic.

## MARL

Matched F1 0.752  
Cooperative learning works when  
prompts align.

## Pipeline

Entity F1 0.616  
beats frontier baselines.

Committee takeaway

**EHR-Copilot improves clinical QA because it separates evidence, reasoning, adversarial verification, and learned debate policy.**



# Architecture alone lifts hallucination detection F1 by 17%.

| Config        | F1    | 95% CI         | Precision | Recall |
|---------------|-------|----------------|-----------|--------|
| Single Critic | 0.551 | [0.522, 0.578] | 0.679     | 0.463  |
| MAD Debate    | 0.645 | [0.622, 0.666] | 0.545     | 0.789  |

**+70%**

Recall lifted from 0.463 → 0.789

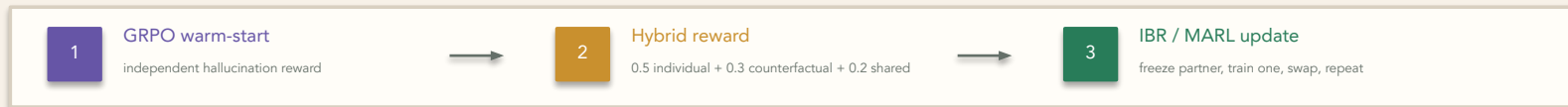
**+0.094**

detection F1 with no training

**Non-overlapping  
confidence  
intervals**

# Cooperative MARL trains debate agents to complement each other — not duplicate.

Three-step flow (matches the right panel of your architecture).



Matched-evaluation training trajectory



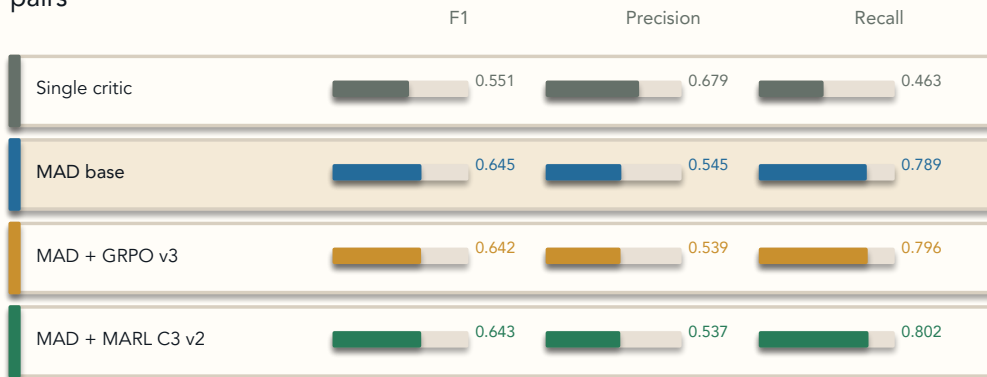
Working recipe

separate LoRA per agent · train one to convergence · swap · repeat

# MAD improves hallucination detection by lifting recall.

The robust improvement comes from the debate structure itself.

Full-pipeline benchmark: 1,000 MedHallu pairs



## Benchmark readout

# +70%

recall lift from single critic  
to MAD base

RL does not improve  
full-pipeline F1 here; the stable  
gain is the debate architecture.

**What it signifies: adversarial challenge catches more hallucinations than a single reviewer.**

# Every layer contributes a different improvement.

4 boxes, one per layer.

## Retrieval

Fine-tuned PubMedBERT

MRR 0.623 → 0.914

## Pipeline

vs frontier LLMs

Entity F1 = 0.616 (beats all)

## Verification (MAD)

Adversarial debate vs single critic

F1 0.551 → 0.645

## Training (MARL)

Cooperative RL with C3

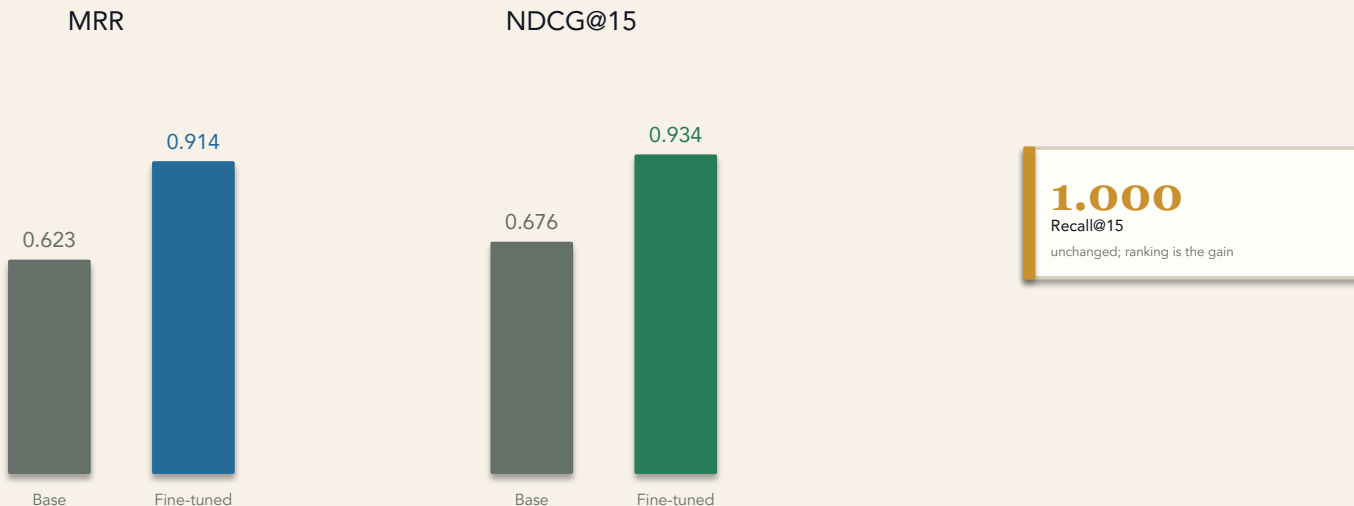
F1 = 0.752

## Takeaway

The system improves because retrieval, pipeline grounding, adversarial verification, and MARL training solve different failure modes.

# Retrieval fine-tuning improves rank quality before generation.

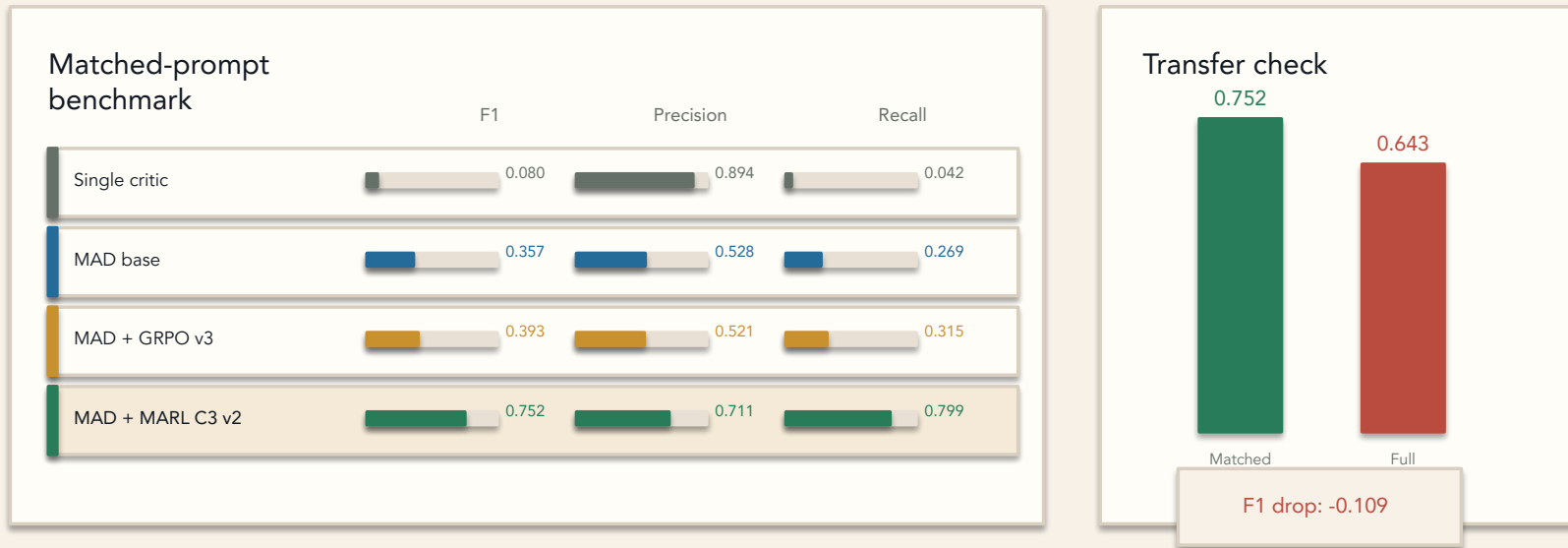
Recall was already saturated; the gain is moving better evidence higher.



**What it signifies: better evidence ordering reduces pressure on the reasoning and debate stages.**

# MARL works when training and deployment prompts match.

The matched-prompt result shows the value of cooperative agent training; the full pipeline exposes transfer mismatch.



**What it signifies: the next research problem is prompt-distribution alignment, not whether MARL can train better agents.**

# Each improvement maps to a different safety control.

The system is not one model trick; it is a layered architecture.

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**Our EHR-System improves clinical QA because it separates evidence, reasoning, adversarial verification, and learned debate policy.**